The refractive index of water, $n_2 = 1.33$

The wavelength of light in water, λ_2

The speed of light in water, υ_2

(a) The equation relating the wavelength and refractive index is,

$$\frac{\lambda_1}{\lambda_2} = \frac{n_2}{n_1}$$

Rewriting, $\lambda_2 = \frac{n_1}{n_2} \times \lambda_1$

Substituting the values,

$$\lambda_2 = \frac{1}{1.33} \times 5893\,\text{Å} = 4431\,\text{Å}$$

$$\lambda_2 = 4431\,\text{Å}$$

(b) The equation relating the speed and refractive index is,

$$\frac{\upsilon_1}{\upsilon_2} = \frac{n_2}{n_1}$$

Rewriting, $\upsilon_2 = \frac{n_1}{n_2} \times \upsilon_1$

Substituting the values,

$$\upsilon_2 = \frac{1}{1.33} \times 3 \times 10^8 = 2.256 \times 10^8$$

$$\upsilon_2 = 2.256 \times 10^8\,\text{m s}^{-1}$$

(c) Frequency of light in vacuum is,

$$\nu_1 = \frac{c}{\lambda_1}$$

Substituting the values,

$$\nu_1 = \frac{3 \times 10^8}{5893 \times 10^{-10}} = 5.091 \times 10^{14}\,\text{Hz}$$

Frequency of light in water is, $\nu_2 = \frac{\nu}{\lambda_2}$

Substituting the values,

$$\nu_2 = \frac{2.256 \times 10^8\,\text{m s}^{-1}}{4431 \times 10^{-10}} = 5.091 \times 10^{14}\,\text{Hz}$$

The results show that the frequency remains same in all media.

7.3 INTERFERENCE

The phenomenon of superposition of two light waves which produces increase in intensity at some points and decrease in intensity at some other points is called *interference* of light.

Superposition of waves refers to addition of waves. The concept of superposition of mechanical waves is studied in (XI Physics 11.7). When two waves simultaneously pass through a particle in a medium, the resultant displacement of that particle is the vector addition of the displacements due to the individual waves. The resultant displacement will be maximum or minimum depending upon the phase difference between the two superimposing waves. These concepts hold good for light as well.

Let us consider two light waves from the two sources S_1 and S_2 meeting at a point P as shown in Figure 7.6.

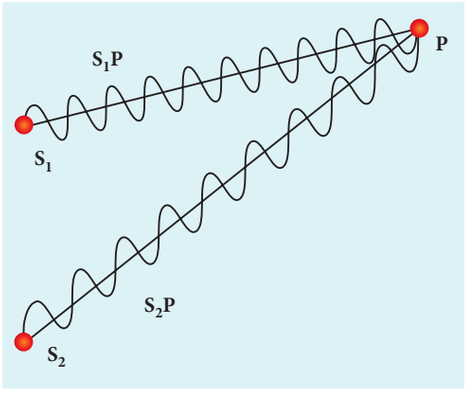


Figure 7.6 Superposition principle

The wave from S_1 at an instant t at P is,

$$y_1 = a_1 \sin \omega t \qquad (7.6)$$

The wave form S_2 at an instant t at P is,

$$y_2 = a_2 \sin (\omega t + \phi) \qquad (7.7)$$

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